

Two-country Version of Hirano, Kishi, and Toda (2025)

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1 Model setup

1.1 Representative household

1.1.1 US

Let

$$S_t \equiv Q_{US,t}n_{US,t} + Q_{W,t}n_{W,t}, \quad \omega_t \equiv \frac{Q_{US,t}n_{US,t}}{S_t}, \quad q_{US,t} \equiv \frac{1}{R_{f,t}}$$

denote, respectively, the market value of equity purchases, the portfolio weight on US equity within equity holdings, and the date- t price of the one-period US riskless bond (gross risk-free return $R_{f,t}$). The US household trades US and RoW equities and the US bond only (it does *not* trade the RoW bond). It chooses

$$\{C_{y,t}, n_{US,t}, n_{W,t}, B_{US,t+1}\}$$

to solve

$$\max (1 - \beta) \log C_{y,t} + \frac{\beta}{1 - \gamma} \log E_t[(C_{o,t+1})^{1-\gamma}] - \frac{\kappa}{2}(\omega_t - \bar{\omega})^2 - \eta \Upsilon(\theta_t)$$

subject to

$$\begin{aligned} C_{y,t} + Q_{US,t}n_{US,t} + Q_{W,t}n_{W,t} + q_{US,t}B_{US,t+1} &= e_{US,t}, \\ C_{o,t+1} &= (Q_{US,t+1} + D_{US,t+1})n_{US,t} + (Q_{W,t+1} + D_{W,t+1})n_{W,t} + B_{US,t+1}. \end{aligned}$$

Here $\bar{\omega} \in [1/2, 1]$ is the targeted portfolio weight on US equity (home bias). The term $\eta \Upsilon(\theta_t)$ captures a convex cost in the present-value bond share θ_t , disciplines large bond positions, and preserves the saving separation. We assume that issuance-cost function Υ satisfies standard regularity and Inada-type conditions.

Assumption 1 (Issuance costs). *The issuance-cost function $\Upsilon : (-\infty, 1) \rightarrow \mathbb{R}$ is C^1 , and satisfies*

$$\lim_{\theta \downarrow -\infty} \Upsilon'(\theta) = -\infty, \quad \Upsilon(\theta) \rightarrow +\infty \text{ as } \theta \downarrow -\infty.$$

Reformulation in terms of total saving, bond share, and equity share. Define the present value of the US bond position and total saving:

$$b_t \equiv q_{US,t} B_{US,t+1}, \quad A_t \equiv S_t + b_t, \quad C_{y,t} = e_{US,t} - A_t.$$

Define the (present-value) bond share in total saving

$$\theta_t \equiv \frac{b_t}{A_t}, \quad S_t = (1 - \theta_t) A_t,$$

and the gross return on total saving

$$R_{A,t+1} \equiv (1 - \theta_t) R_{p,t+1} + \theta_t R_{f,t}, \quad R_{f,t} \equiv \frac{1}{q_{US,t}},$$

where

$$R_{p,t+1} = \omega_t R_{US,t+1} + (1 - \omega_t) R_{W,t+1}, \quad R_{i,t+1} = \frac{Q_{i,t+1} + D_{i,t+1}}{Q_{i,t}}, \quad i \in \{US, W\}.$$

Old-age consumption satisfies $C_{o,t+1} = A_t R_{A,t+1}$, so the problem can be written as

$$\max_{A_t, \omega_t, \theta_t} (1 - \beta) \log(e_{US,t} - A_t) + \frac{\beta}{1 - \gamma} \log E_t [(A_t R_{A,t+1})^{1-\gamma}] - \frac{\kappa}{2} (\omega_t - \bar{\omega})^2 - \eta \Upsilon(\theta_t).$$

Under log utility in youth and the risk-sensitive aggregator in old age, the optimal saving choice yields

$$\frac{1 - \beta}{e_{US,t} - A_t} = \frac{\beta}{A_t} \Rightarrow A_t = \beta e_{US,t}, \quad C_{y,t} = (1 - \beta) e_{US,t}. \quad (1)$$

Portfolio FOCs and stock-pricing implications. Define the normalized *undis-*
torted portfolio kernel

$$\mathcal{K}_{t,t+1} \equiv \frac{R_{A,t+1}^{-\gamma}}{E_t [R_{A,t+1}^{1-\gamma}]}, \quad E_t [\mathcal{K}_{t,t+1} R_{A,t+1}] = 1. \quad (2)$$

The FOC for ω_t implies

$$\beta(1 - \theta_t) E_t [\mathcal{K}_{t,t+1} (R_{US,t+1} - R_{W,t+1})] = \kappa(\omega_t - \bar{\omega}). \quad (3)$$

The FOC for θ_t implies

$$\beta E_t [\mathcal{K}_{t,t+1} (R_{f,t} - R_{p,t+1})] = \eta \Upsilon'(\theta_t). \quad (4)$$

For later use, define

$$\phi_t \equiv \frac{\kappa}{\beta(1-\theta_t)}(\omega_t - \bar{\omega}) = \frac{\kappa C_{y,t}}{(1-\beta)S_t}(\omega_t - \bar{\omega}), \quad \lambda_t \equiv \frac{\eta\theta_t}{\beta} \Upsilon'(\theta_t).$$

Using

$$R_{A,t+1} = (1 - \theta_t)R_{p,t+1} + \theta_t R_{f,t}$$

and (2), the bond-share FOC implies

$$E_t[\mathcal{K}_{t,t+1} R_{p,t+1}] = 1 - \lambda_t. \quad (5)$$

Combining (5) with (3) yields

$$E_t[\mathcal{K}_{t,t+1} R_{US,t+1}] = 1 - \lambda_t + \phi_t(1 - \omega_t), \quad (6)$$

$$E_t[\mathcal{K}_{t,t+1} R_{W,t+1}] = 1 - \lambda_t - \phi_t \omega_t. \quad (7)$$

Therefore the stock-pricing equations can be written as

$$Q_{US,t} = E_t \left[\widetilde{M}_{t,t+1}^{US} (Q_{US,t+1} + D_{US,t+1}) \right], \quad (8)$$

$$Q_{W,t} = E_t \left[\widetilde{M}_{t,t+1}^W (Q_{W,t+1} + D_{W,t+1}) \right],$$

with

$$\begin{aligned} \widetilde{M}_{t,t+1}^{US} &= \frac{\mathcal{K}_{t,t+1}}{1 - \lambda_t + \phi_t(1 - \omega_t)}, \\ \widetilde{M}_{t,t+1}^W &= \frac{\mathcal{K}_{t,t+1}}{1 - \lambda_t - \phi_t \omega_t}. \end{aligned}$$

Thus the kernel that prices the US stock differs from the undistorted kernel $\mathcal{K}_{t,t+1}$ because of both the home-bias wedge ϕ_t and the bond-share wedge λ_t .

1.1.2 Rest-of-World (RoW)

Let

$$S_t^* \equiv Q_{US,t} n_{US,t}^* + Q_{W,t} n_{W,t}^*, \quad \omega_t^* \equiv \frac{Q_{US,t} n_{US,t}^*}{S_t^*}, \quad q_{W,t} \equiv \frac{1}{R_{f,t}^W}$$

where $q_{W,t}$ is the date- t price of the one-period RoW riskless bond (gross return $R_{f,t}^W$). The RoW household trades both equities, the US bond, and the RoW bond, and chooses

$$\{C_{y,t}^*, n_{US,t}^*, n_{W,t}^*, B_{US,t+1}^*, B_{W,t+1}^*\}$$

to solve

$$\max U_t^* = (1 - \beta) \log C_{y,t}^* + \frac{\beta}{1 - \gamma} \log E_t \left[(C_{o,t+1}^*)^{1-\gamma} \right] - \frac{\kappa}{2} (\omega_t^* - \bar{\omega}^*)^2 + \chi \log(q_{US,t} B_{US,t+1}^*),$$

subject to

$$\begin{aligned} C_{y,t}^* + Q_{US,t} n_{US,t}^* + Q_{W,t} n_{W,t}^* + q_{US,t} B_{US,t+1}^* + q_{W,t} B_{W,t+1}^* &= e_{W,t}, \\ C_{o,t+1}^* &= (Q_{US,t+1} + D_{US,t+1}) n_{US,t}^* + (Q_{W,t+1} + D_{W,t+1}) n_{W,t}^* + B_{US,t+1}^* + B_{W,t+1}^*. \end{aligned}$$

The term $\chi > 0$ generates a convenience yield from holding US bonds and the “exorbitant privilege” of US bonds as a consequence.

Stock pricing (RoW effective SDF). The FOCs for equity positions imply, for $i \in \{US, W\}$,

$$Q_{i,t} = E_t \left[\tilde{M}_{t,t+1}^{*,i} (Q_{i,t+1} + D_{i,t+1}) \right],$$

with

$$\begin{aligned} \tilde{M}_{t,t+1}^{*,US} &= \frac{M_{t,t+1}^*}{1 + \phi_t^* (1 - \omega_t^*)}, & \tilde{M}_{t,t+1}^{*,W} &= \frac{M_{t,t+1}^*}{1 - \phi_t^* \omega_t^*}, \\ M_{t,t+1}^* &= \frac{\beta}{1 - \beta} \frac{C_{y,t}^* (C_{o,t+1}^*)^{-\gamma}}{E_t [(C_{o,t+1}^*)^{1-\gamma}]}, & \phi_t^* &\equiv \frac{\kappa C_{y,t}^*}{(1 - \beta) S_t^*} (\omega_t^* - \bar{\omega}^*). \end{aligned}$$

Saving notation and portfolio shares. Define present values and total saving:

$$b_{US,t}^* \equiv q_{US,t} B_{US,t+1}^*, \quad b_{W,t}^* \equiv q_{W,t} B_{W,t+1}^*, \quad A_t^* \equiv S_t^* + b_{US,t}^* + b_{W,t}^*, \quad C_{y,t}^* = e_{W,t} - A_t^*.$$

Define bond shares in total saving:

$$\theta_{US,t}^* \equiv \frac{b_{US,t}^*}{A_t^*}, \quad \theta_{W,t}^* \equiv \frac{b_{W,t}^*}{A_t^*}, \quad S_t^* = (1 - \theta_{US,t}^* - \theta_{W,t}^*) A_t^*.$$

Define the gross return on total saving:

$$R_{A,t+1}^* \equiv (1 - \theta_{US,t}^* - \theta_{W,t}^*) R_{p,t+1}^* + \theta_{US,t}^* R_{f,t} + \theta_{W,t}^* R_{f,t}^W,$$

where

$$R_{f,t} \equiv \frac{1}{q_{US,t}}, \quad R_{f,t}^W \equiv \frac{1}{q_{W,t}}, \quad R_{p,t+1}^* = \omega_t^* R_{US,t+1} + (1 - \omega_t^*) R_{W,t+1}.$$

Then $C_{o,t+1}^* = A_t^* R_{A,t+1}^*$.

RoW saving rule. Because $\chi \log(q_{US,t} B_{US,t+1}^*) = \chi \log(\theta_{US,t}^* A_t^*)$ contributes $\chi \log A_t^*$, the saving choice separates and yields

$$\frac{1 - \beta}{e_{W,t} - A_t^*} = \frac{\beta + \chi}{A_t^*} \Rightarrow A_t^* = \frac{\beta + \chi}{1 + \chi} e_{W,t}, \quad C_{y,t}^* = \frac{1 - \beta}{1 + \chi} e_{W,t}. \quad (9)$$

RoW portfolio FOCs. Define the normalized RoW portfolio kernel

$$\mathcal{M}_{t,t+1}^* \equiv \frac{(R_{A,t+1}^*)^{-\gamma}}{E_t[(R_{A,t+1}^*)^{1-\gamma}]}.$$

The FOC for ω_t^* is

$$\beta(1 - \theta_{US,t}^* - \theta_{W,t}^*) E_t[\mathcal{M}_{t,t+1}^* (R_{US,t+1} - R_{W,t+1})] = \kappa(\omega_t^* - \bar{\omega}^*). \quad (10)$$

The FOC for the RoW bond share $\theta_{W,t}^*$ is

$$\beta E_t[\mathcal{M}_{t,t+1}^* (R_{f,t}^W - R_{p,t+1}^*)] = 0. \quad (11)$$

The FOC for the US bond share $\theta_{US,t}^*$ is

$$\beta E_t[\mathcal{M}_{t,t+1}^* (R_{f,t} - R_{p,t+1}^*)] + \frac{\chi}{\theta_{US,t}^*} = 0. \quad (12)$$

And the FOCs for the bond amounts $(B_{W,t+1}^*, B_{US,t+1}^*)$ imply the bond-pricing equations for RoW agents:

$$q_{W,t} = E_t[M_{t,t+1}^*], \quad (13)$$

$$q_{US,t} = E_t[M_{t,t+1}^*] + \frac{\chi}{1 + \chi} \frac{e_{W,t}}{B_{US,t+1}^*}. \quad (14)$$

Since the convenience-yield term $\chi \log(q_{US,t} B_{US,t+1}^*)$ requires $B_{US,t+1}^* > 0$, the wedge term in (14) is strictly positive; combining with (13) immediately gives $q_{US,t} > q_{W,t}$, generating the exorbitant privilege of US bonds.

1.2 Market clearing

World stock market clearing.

$$\begin{aligned} n_{US,t} + n_{US,t}^* &= 1, \\ n_{W,t} + n_{W,t}^* &= 1. \end{aligned} \quad (15)$$

Bond market clearing. Assume both one-period riskless bonds are in zero net supply. Since only RoW trades the RoW bond, bond market clearing is

$$B_{US,t+1} + B_{US,t+1}^* = 0, \quad B_{W,t+1}^* = 0. \quad (16)$$

Equivalently, in present values,

$$b_t + b_{US,t}^* = 0, \quad b_{W,t}^* = 0.$$

Stock market clearing in value form. Using portfolio weights,

$$\begin{aligned} Q_{US,t} &= \omega_t S_t + \omega_t^* S_t^*, \\ Q_{W,t} &= (1 - \omega_t) S_t + (1 - \omega_t^*) S_t^*. \end{aligned} \quad (17)$$

Aggregate market-cap identity. Since $S_t = A_t - b_t$ and $S_t^* = A_t^* - b_{US,t}^* - b_{W,t}^*$, we have

$$Q_{US,t} + Q_{W,t} = S_t + S_t^* = A_t + A_t^* - (b_t + b_{US,t}^*) - b_{W,t}^*.$$

Using bond clearing (16) and the saving rules (1) and (9), it follows that

$$\boxed{Q_{US,t} + Q_{W,t} = \beta e_{US,t} + \frac{\beta + \chi}{1 + \chi} e_{W,t}.} \quad (18)$$

Goods market clearing.

$$C_{y,t} + C_{y,t}^* + C_{o,t} + C_{o,t}^* = e_{US,t} + e_{W,t} + D_{US,t} + D_{W,t}. \quad (19)$$

2 Existence of bubbles

In this section we reserve $\mathcal{K}_{t,t+1}$ for the undistorted normalized portfolio kernel,

$$\mathcal{K}_{t,t+1} = \frac{R_{A,t+1}^{-\gamma}}{E_t[R_{A,t+1}^{1-\gamma}]},$$

and we let $\mathcal{M}_{t,t+1}$ denote the *effective* kernel that prices the US stock. By (8),

$$\mathcal{M}_{t,t+1} \equiv \widetilde{M}_{t,t+1}^{US} = \frac{\mathcal{K}_{t,t+1}}{\Psi_t}, \quad \Psi_t \equiv 1 - \lambda_t + \phi_t(1 - \omega_t), \quad (20)$$

where

$$\phi_t = \frac{\kappa}{\beta(1 - \theta_t)}(\omega_t - \bar{\omega}), \quad \lambda_t = \frac{\eta\theta_t}{\beta} \Upsilon'(\theta_t).$$

By (1) and the identity $C_{o,t+1} = A_t R_{A,t+1}$, the undistorted intertemporal marginal rate of substitution equals

$$\frac{\beta}{1-\beta} \frac{C_{y,t} C_{o,t+1}^{-\gamma}}{E_t[C_{o,t+1}^{1-\gamma}]} = \mathcal{K}_{t,t+1}.$$

Hence the only difference between the stock-pricing kernel $\mathcal{M}_{t,t+1}$ and the undistorted kernel $\mathcal{K}_{t,t+1}$ is the time- t portfolio wedge Ψ_t^{-1} .

We impose the following regularity condition.

Assumption 2 (Regular effective kernel).

$$\Psi_t > 0 \quad a.s. \text{ for all } t.$$

Also note that because the RoW utility includes $\chi \log(q_{US,t} B_{US,t+1}^*)$, feasibility requires $B_{US,t+1}^* > 0$. With zero net supply of US bonds, $B_{US,t+1} + B_{US,t+1}^* = 0$ implies $B_{US,t+1} < 0$, hence

$$\theta_t = \frac{b_t}{A_t} < 0 \quad \Rightarrow \quad 1 - \theta_t > 1. \quad (21)$$

Then we show that $\Psi_t > 0$ are very likely to hold under mild parameter restrictions. Because $0 \leq \omega_t \leq 1$, we have $1 - \omega_t \in [0, 1]$ and

$$(\omega_t - \bar{\omega})(1 - \omega_t) \geq -\bar{\omega}(1 - \omega_t) \geq -\bar{\omega}.$$

Therefore

$$\phi_t(1 - \omega_t) = \frac{\kappa}{\beta(1 - \theta_t)}(\omega_t - \bar{\omega})(1 - \omega_t) \geq -\frac{\kappa\bar{\omega}}{\beta(1 - \theta_t)}.$$

We further assume the function form of issuance costs of bonds, for now, as $\Upsilon(\theta_t) = -\log(1 - \theta_t) + \theta_t^2/2$. Then $\Psi_t > 0$ if

$$\beta > \frac{\eta\theta_t}{1 - \theta_t} + \eta\theta_t^2 + \frac{\kappa\bar{\omega}}{1 - \theta_t}, \quad (22)$$

as

$$\Psi_t > 1 - \frac{\eta\theta_t}{\beta} \Upsilon'(\theta_t) - \frac{\kappa\bar{\omega}}{\beta(1 - \theta_t)}.$$

It is immediate that (22) easily hold for sufficiently high β and sufficiently low η and κ as θ_t usually falls within $[-1, 0)$ and $1 - \theta_t > 1$.

2.1 Temporary unbalanced growth and stochastic bubbles

Following [Hirano et al. \(2025\)](#), let

$$\mathcal{M}_{t,t+s} := \prod_{j=0}^{s-1} \mathcal{M}_{t+j,t+j+1}.$$

The US stock pricing equation (8) (for $i = US$) implies

$$Q_{US,0} = \mathbb{E}_0 \left[\sum_{s=1}^{\infty} \mathcal{M}_{0,s} D_{US,s} \right] + \lim_{t \rightarrow \infty} \mathbb{E}_0 [\mathcal{M}_{0,t} Q_{US,t}],$$

or more generally,

$$Q_{US,t} = \underbrace{\mathbb{E}_t \left[\sum_{s=1}^{\infty} \mathcal{M}_{t,t+s} D_{US,t+s} \right]}_{\text{fundamental value } V_{US,t}} + \underbrace{\lim_{T \rightarrow \infty} \mathbb{E}_t [\mathcal{M}_{t,T} Q_{US,T}]}_{\text{bubble } B_{US,t}}.$$

Consider a regime-switching model with an absorbing state, as in [Weil \(1987\)](#).

Assumption 3. *There are two states of the US economy denoted by u, b . Letting $z_t \in \{u, b\}$ denote the state at time t , the transition probabilities are given by*

$$P[z_{t+1} = u \mid z_t = u] = \pi \in (0, 1), \quad P[z_{t+1} = b \mid z_t = b] = 1.$$

Assumption 3 implies that state b is absorbing. We further assume that once state b is reached, uncertainty is resolved, and endowments and dividends grow at the same rate (balanced growth). On the other hand, the RoW is always on its balanced growth path (BGP hereafter).

Assumption 4. *For any τ , conditional on $z_\tau = b$, the sequence $\{e_{US,t}, D_{US,t}, e_{W,t}, D_{W,t}\}_{t=\tau}^{\infty}$ is deterministic and*

$$\frac{e_{US,t+1}}{e_{US,t}} = \frac{D_{US,t+1}}{D_{US,t}} = \frac{e_{W,t+1}}{e_{W,t}} = \frac{D_{W,t+1}}{D_{W,t}}, \quad \forall t \geq \tau.$$

We can show that once state b is reached, there is no bubble in the US stock market.

Proposition 1. *If **Assumptions 3** and **4** hold, once state b is reached, the US stock price equals its fundamental value: $Q_{US,t} = V_{US,t}$.*

Proof. The proof directly follows from [Hirano et al. \(2025\)](#) because the US stock continues to satisfy a standard Euler equation with the effective kernel $\mathcal{M}_{t,t+1}$, and state b is deterministic balanced growth. $\square$

By [Proposition 1](#), bubbles cannot arise in state b . So we assume $z_0 = u$ from now on. Let τ be the stopping time that represents the first time the US economy switches to state b : $\tau = \min\{t : z_t = b\}$. By [Assumption 3](#), τ follows a geometric distribution,

$$P[\tau = j] = \pi^{j-1}(1 - \pi), \quad j = 1, 2, \dots$$

Then the bubble term can be written as

$$E_0 [\mathcal{M}_{0,t} Q_{US,t}] = \sum_{j=1}^t \pi^{j-1}(1 - \pi) E_0 [\mathcal{M}_{0,t} Q_{US,t} \mid \tau = j] + \pi^t E_0 [\mathcal{M}_{0,t} Q_{US,t} \mid \tau > t].$$

The following proposition provides a sufficient condition for the nonexistence of bubbles.

Proposition 2. Suppose [Assumptions 3](#) and [4](#) hold. Then there is no bubble if and only if

$$\lim_{t \rightarrow \infty} \pi^t E_0 [\mathcal{M}_{0,t} Q_{US,t} \mid \tau > t] = 0.$$

If $(e_{US,t}, D_{US,t})$ is known at time $t - 1$ (i.e., $(e_{US,t}, D_{US,t})$ is $\mathcal{F}_{t-1}$ -measurable), then there is no bubble.

Proof. The proof directly follows from [Hirano et al. \(2025\)](#). □

Aligned with the sufficient conditions in [Proposition 2](#), we introduce the following assumption.

Assumption 5. Conditional on time $t - 1$ information, the US endowment $e_{US,t}$ and US dividend $D_{US,t}$ depend only on the state $z_t \in \{u, b\}$.

Now we are ready to present a more general but less primitive version of Theorem 1 in [Hirano et al. \(2025\)](#). For $z \in \{u, b\}$, let x_t^z be the value of $x_{US,t}^z$ conditional on $z_0 = \dots = z_{t-1} = u$ and $z_t = z$, and let $C_t^z = C_{o,t}^z$ denote US old-age consumption at date t on branch z .

Theorem 1. Suppose [Assumptions 3](#), [4](#), and [5](#) hold. If $z_0 = u$, then there is a bubble at $t = 0$ if and only if

$$\sum_{t=1}^{\infty} \frac{D_t^u}{Q_t^u} < \infty, \tag{23a}$$

$$\sum_{t=1}^{\infty} \frac{Q_t^b + D_t^b}{(C_t^b)^\gamma} / \left(\frac{Q_t^u}{(C_t^u)^\gamma} \right) < \infty. \tag{23b}$$

Proof. Let

$$\mathcal{B}_t \equiv E_0 [\mathcal{M}_{0,t} Q_t^u \mathbf{1}\{\tau > t\}].$$

Because $\mathbb{P}(\tau > t) = \pi^t$, this is equivalent to

$$\mathcal{B}_t = \pi^t \mathbb{E}_0[\mathcal{M}_{0,t} Q_t^u \mid \tau > t].$$

By [Proposition 2](#), there is a bubble at $t = 0$ if and only if

$$\lim_{t \rightarrow \infty} \mathcal{B}_t > 0,$$

and there is no bubble if and only if $\lim_{t \rightarrow \infty} \mathcal{B}_t = 0$.

On histories with $z_{t-1} = u$, the one-step US stock pricing equation is

$$Q_{t-1}^u = \mathbb{E}_{t-1} \left[\mathcal{M}_{t-1,t} \left((Q_t^u + D_t^u) \mathbf{1}_{\{z_t=u\}} + (Q_t^b + D_t^b) \mathbf{1}_{\{z_t=b\}} \right) \right].$$

Multiply both sides by $\mathcal{M}_{0,t-1} \mathbf{1}_{\{\tau > t-1\}}$ and take $\mathbb{E}_0$. Using $\mathcal{M}_{0,t} = \mathcal{M}_{0,t-1} \mathcal{M}_{t-1,t}$, we obtain

$$\mathbb{E}_0 [\mathcal{M}_{0,t-1} Q_{t-1}^u \mathbf{1}_{\{\tau > t-1\}}] = \mathbb{E}_0 [\mathcal{M}_{0,t} (Q_t^u + D_t^u) \mathbf{1}_{\{\tau > t\}}] + \mathbb{E}_0 [\mathcal{M}_{0,t} (Q_t^b + D_t^b) \mathbf{1}_{\{\tau = t\}}]. \quad (24)$$

Since

$$\mathbb{E}_0 [\mathcal{M}_{0,t} (Q_t^u + D_t^u) \mathbf{1}_{\{\tau > t\}}] = \mathcal{B}_t + \mathbb{E}_0 [\mathcal{M}_{0,t} D_t^u \mathbf{1}_{\{\tau > t\}}],$$

(24) becomes

$$\mathcal{B}_{t-1} = \mathcal{B}_t + \underbrace{\mathbb{E}_0 [\mathcal{M}_{0,t} D_t^u \mathbf{1}_{\{\tau > t\}}]}_{\text{dividend leakage term}} + \underbrace{\mathbb{E}_0 [\mathcal{M}_{0,t} (Q_t^b + D_t^b) \mathbf{1}_{\{\tau = t\}}]}_{\text{switch-at-t leakage term}}, \quad (25)$$

where the last two terms can be interpreted as leakage terms that prevent the bubble from growing too fast. In other words, *leakage* is defined as the amount of present value that drains out of the continuation bubble mass when we move one period forward. It's exactly the gap $\mathcal{B}_{t-1} - \mathcal{B}_t$.

Define the leakage ratio

$$a_t \equiv \frac{\mathbb{E}_0 [\mathcal{M}_{0,t} D_t^u \mathbf{1}_{\{\tau > t\}}] + \mathbb{E}_0 [\mathcal{M}_{0,t} (Q_t^b + D_t^b) \mathbf{1}_{\{\tau = t\}}]}{\mathcal{B}_t}.$$

Then (25) implies

$$\mathcal{B}_{t-1} = (1 + a_t) \mathcal{B}_t,$$

so iterating yields

$$\mathcal{B}_t = \mathcal{B}_0 \prod_{s=1}^t \frac{1}{1 + a_s}, \quad \mathcal{B}_0 = Q_0^u > 0.$$

Since $a_s \geq 0$,

$$\mathcal{B}_0 \left(1 + \sum_{s=1}^t a_s \right)^{-1} \geq \mathcal{B}_t \geq \mathcal{B}_0 \exp \left(- \sum_{s=1}^t a_s \right).$$

Hence

$$\lim_{t \rightarrow \infty} \mathcal{B}_t > 0 \iff \sum_{t=1}^{\infty} a_t < \infty.$$

It remains to express a_t in terms of branch variables. On $\{\tau > t\}$ we have $z_1 = \dots = z_t = u$, so

$$\frac{\mathbb{E}_0[\mathcal{M}_{0,t} D_t^u \mathbf{1}\{\tau > t\}]}{\mathcal{B}_t} = \frac{D_t^u}{Q_t^u}.$$

On $\{\tau = t\}$ we have $z_1 = \dots = z_{t-1} = u$ and $z_t = b$, hence

$$\mathbb{E}_0[\mathcal{M}_{0,t}(Q_t^b + D_t^b) \mathbf{1}\{\tau = t\}] = \pi^{t-1} (1 - \pi) \mathcal{M}_{0,t-1}^u \mathcal{M}_{t-1,t}^b (Q_t^b + D_t^b),$$

while

$$\mathcal{B}_t = \pi^t \mathcal{M}_{0,t-1}^u \mathcal{M}_{t-1,t}^u Q_t^u.$$

Therefore,

$$\frac{\mathbb{E}_0[\mathcal{M}_{0,t}(Q_t^b + D_t^b) \mathbf{1}\{\tau = t\}]}{\mathcal{B}_t} = \frac{1 - \pi}{\pi} \frac{\mathcal{M}_{t-1,t}^b}{\mathcal{M}_{t-1,t}^u} \frac{Q_t^b + D_t^b}{Q_t^u},$$

which is the ratio of the discounted b-state cum-dividend stock price to the discounted u-state ex-dividend stock price.

Finally, under **Assumption 5**, the time- $(t-1)$ portfolio choices $(\omega_{t-1}, \theta_{t-1})$ are $\mathcal{F}_{t-1}$ -measurable and do not depend on z_t . Hence the time- $(t-1)$ wedge

$$\Psi_{t-1} = 1 - \lambda_{t-1} + \phi_{t-1}(1 - \omega_{t-1})$$

is common across the two realizations $z_t \in \{u, b\}$, so it cancels in the ratio:

$$\frac{\mathcal{M}_{t-1,t}^b}{\mathcal{M}_{t-1,t}^u} = \frac{\mathcal{K}_{t-1,t}^b}{\mathcal{K}_{t-1,t}^u}.$$

Moreover, the denominator $E_{t-1}[R_{A,t}^{1-\gamma}]$ in $\mathcal{K}_{t-1,t}$ is also common across the two realizations, so

$$\frac{\mathcal{K}_{t-1,t}^b}{\mathcal{K}_{t-1,t}^u} = \left(\frac{R_{A,t}^b}{R_{A,t}^u} \right)^{-\gamma} = \left(\frac{C_t^b}{C_t^u} \right)^{-\gamma}$$

Therefore,

$$a_t = \frac{D_t^u}{Q_t^u} + \frac{1 - \pi}{\pi} \left(\frac{C_t^u}{C_t^b} \right)^\gamma \frac{Q_t^b + D_t^b}{Q_t^u} = \frac{D_t^u}{Q_t^u} + \frac{1 - \pi}{\pi} \frac{Q_t^b + D_t^b}{(C_t^b)^\gamma} \frac{(C_t^b)^\gamma}{Q_t^u}.$$

Hence $\sum_{t=1}^{\infty} a_t < \infty$ is equivalent to (23). $\square$

Intuitively, the first summability condition (23a) requires that dividends along the u -history do not grow too fast relative to stock prices, while the second summability condition (23b) requires that the stock prices discounted by the consumption-based SDF along the u -history grow much faster than the cum-dividend stock prices would have in b -state. Therefore, the bubble candidate survives if and only if, over time, dividends and switching events do not drain too much value relative to what remains; that is, if and only if the cumulative leakage $\sum a_t$ is finite.

2.2 Primitive sufficient and necessary conditions

We now derive primitive conditions for the existence of bubbles by translating the two summability conditions in Theorem 1 into exogenous variables along the u -history and at the switching date. We first impose primitive boundedness restrictions on relative country size, switch-date US fundamentals, portfolio weights, and continuation-branch growth and dividends.

Assumption 6 (Relative country size and dividend bounds). *There exists $\bar{\lambda}_{ew} < \infty$ and $\bar{\lambda}_{D_W} < \infty$ such that*

$$\frac{e_{W,t}}{e_{US,t}^z} \leq \bar{\lambda}_{ew}, \quad \frac{D_{W,t}}{e_{US,t}^z} \leq \bar{\lambda}_{D_W}, \quad \text{for all } t \leq \tau.$$

Assumption 7 (Switch-date US fundamentals). *There exist constants $0 < \bar{\lambda}_e < \infty$ and $0 \leq \bar{\lambda}_{D_{US}^b} < \infty$ such that, whenever $z_{t-1} = u$ and $z_t = b$,*

$$\frac{e_{US,t}^b}{e_{US,t}^u} \leq \bar{\lambda}_e, \quad \frac{D_{US,t}^b}{e_{US,t}^u} \leq \bar{\lambda}_{D_{US}^b}.$$

Assumption 8 (Uniformly interior equity weights). *There exists $\underline{\lambda}_\omega \in (0, 1/2)$ such that, for each $z \in \{u, b\}$ and all $t \leq \tau$,*

$$\underline{\lambda}_\omega \leq \omega_t^z, \quad \omega_t^{*,z} \leq 1 - \underline{\lambda}_\omega.$$

Assumption 9 (Continuation-branch US dividends and one-step US growth). *There exist constants $0 < \underline{g}_u \leq \bar{g}_u < \infty$ and $\bar{\lambda}_{D_{US}} < \infty$ such that*

$$\underline{g}_u \leq \frac{e_{US,t}^u}{e_{US,t-1}^u} \leq \bar{g}_u, \quad \frac{D_{US,t}^z}{e_{US,t}^z} \leq \bar{\lambda}_{D_{US}}, \quad t \geq 1.$$

Then in the following lemmas, we show that stock prices are bounded and proportional to endowments; thus, the return rates for stocks, bonds, and the mixed portfolio are also finite, which finally leads to consumption that is proportional to endowments.

Lemma 1 (Equilibrium bounds on stock market values). *Suppose **Assumptions 6** and **8** hold. Then there exist constants $0 < \underline{q} \leq \bar{q} < \infty$ such that, for each $i \in \{US, W\}$, each $z \in \{u, b\}$, and all $t \leq \tau$,*

$$\underline{q} e_{US,t}^z \leq Q_{i,t}^z \leq \bar{q} e_{US,t}^z,$$

where one may take

$$\underline{q} \equiv \underline{\lambda}_\omega \beta, \quad \bar{q} \equiv \beta + \frac{\beta + \chi}{1 + \chi} \bar{\lambda}_{e_W}.$$

Proof. For $z = u$, **(1)**, **(17)**, **(21)**, and **Assumption 8** imply

$$Q_{US,t}^u = \omega_t^u S_t^u + \omega_t^{*,u} S_t^{*,u} \geq \omega_t^u S_t^u \geq \underline{\lambda}_\omega (1 - \theta_t^u) A_t^u \geq \underline{\lambda}_\omega \beta e_{US,t}^u,$$

and similarly

$$Q_{W,t}^u = (1 - \omega_t^u) S_t^u + (1 - \omega_t^{*,u}) S_t^{*,u} \geq (1 - \omega_t^u) S_t^u \geq \underline{\lambda}_\omega \beta e_{US,t}^u.$$

For $z = b$,

$$Q_{US,t}^b \geq \omega_t^b S_t^b \geq \underline{\lambda}_\omega (1 - \theta_t^b) A_t^b \geq \underline{\lambda}_\omega \beta e_{US,t}^b,$$

and likewise

$$Q_{W,t}^b \geq (1 - \omega_t^b) S_t^b \geq \underline{\lambda}_\omega \beta e_{US,t}^b.$$

Hence

$$Q_{i,t}^z \geq \underline{q} e_{US,t}^z, \quad \underline{q} \equiv \underline{\lambda}_\omega \beta.$$

For the upper bound, **(18)**, **Assumption 6** imply

$$Q_{i,t}^z \leq Q_{US,t}^z + Q_{W,t}^z = \beta e_{US,t}^z + \frac{\beta + \chi}{1 + \chi} e_{W,t} \leq \left(\beta + \frac{\beta + \chi}{1 + \chi} \bar{\lambda}_{e_W} \right) e_{US,t}^z.$$

Thus

$$Q_{i,t}^z \leq \bar{q} e_{US,t}^z, \quad \bar{q} \equiv \beta + \frac{\beta + \chi}{1 + \chi} \bar{\lambda}_{ew}.$$

□

Lemma 2 (Equilibrium bounds on risky returns). *Suppose **Assumptions 6 to 9** hold, and suppose dividends are nonnegative:*

$$D_{US,t} \geq 0, \quad D_{W,t} \geq 0, \quad \forall t.$$

Then there exist constants $0 < \bar{R} < \infty$ such that, for each $z \in \{u, b\}$ and all $t \leq \tau$,

$$0 < R_{US,t}^z, R_{W,t}^z, R_{p,t}^z, R_{p,t}^{*,z} \leq \bar{R}^z.$$

Proof. By **Lemma 1**,

$$\underline{q} e_{US,t}^z \leq Q_{i,t}^z \leq \bar{q} e_{US,t}^z, \quad i \in \{US, W\}, \quad z \in \{u, b\}.$$

Let

$$\bar{d}_{US} \equiv \max\{\bar{\lambda}_{D_{US}}, \bar{\lambda}_{D_{US}^b}\}, \quad \bar{d}_W \equiv \bar{\lambda}_{D_W}, \quad \bar{d} \equiv \max\{\bar{d}_{US}, \bar{d}_W\}.$$

Then

$$D_{US,t}^z \leq \bar{d}_{US} e_{US,t}^u, \quad D_{W,t}^z \leq \bar{d}_W e_{US,t}^u, \quad z \in \{u, b\}, \quad \forall t \leq \tau.$$

For $i \in \{US, W\}$ and $z \in \{u, b\}$, the one-period stock return from date $t - 1$ on the u -history to date t on branch z is

$$R_{i,t}^z = \frac{Q_{i,t}^z + D_{i,t}^z}{Q_{i,t-1}^u}.$$

Since dividends are nonnegative,

$$\begin{aligned} R_{i,t}^u &\geq \frac{Q_{i,t}^u}{Q_{i,t-1}^u} \geq \frac{\underline{q} e_{US,t}^u}{\bar{q} e_{US,t-1}^u} \geq \frac{\underline{q} \underline{g}_u}{\bar{q}} \equiv \underline{R}^u > 0. \\ R_{i,t}^b &\geq \frac{Q_{i,t}^b}{Q_{i,t-1}^u} \geq \frac{\underline{q} e_{US,t}^b}{\bar{q} e_{US,t-1}^u} \equiv \underline{R}_t^b > 0. \end{aligned}$$

Similarly,

$$\begin{aligned} R_{i,t}^u &\leq \frac{Q_{i,t}^u + D_{i,t}^u}{Q_{i,t-1}^u} \leq \frac{(\bar{q} + \bar{d}) e_{US,t}^u}{\underline{q} e_{US,t-1}^u} \leq \frac{(\bar{q} + \bar{d}) \bar{g}_u}{\underline{q}} \equiv \bar{R}^u < \infty. \\ R_{i,t}^b &\leq \frac{Q_{i,t}^b + D_{i,t}^b}{Q_{i,t-1}^u} \leq \frac{(\bar{q} + \bar{d}) e_{US,t}^b}{\underline{q} e_{US,t-1}^u} \leq \frac{(\bar{q} + \bar{d}) \bar{g}_u \bar{\lambda}_e}{\underline{q}} \equiv \bar{R}^b < \infty. \end{aligned}$$

Now

$$R_{p,t}^z = \omega_{t-1}^u R_{US,t}^z + (1 - \omega_{t-1}^u) R_{W,t}^z,$$

and

$$R_{p,t}^{*,z} = \omega_{t-1}^{*,u} R_{US,t}^z + (1 - \omega_{t-1}^{*,u}) R_{W,t}^z.$$

Because **Assumption 8** implies

$$0 \leq \omega_{t-1}^u, \omega_{t-1}^{*,u} \leq 1,$$

both $R_{p,t}^z$ and $R_{p,t}^{*,z}$ are convex combinations of $R_{US,t}^z$ and $R_{W,t}^z$. Therefore

$$0 < R_{p,t}^z, R_{p,t}^{*,z} \leq \bar{R}^z.$$

□

Lemma 3 (Equilibrium upper bound on the US risk-free rate). *Under the assumptions of **Lemma 2**, there exists $\bar{R}_f < \infty$ such that*

$$R_{f,t} \leq \bar{R}_f \quad \text{for all } t \leq \tau.$$

One may take

$$\bar{R}_f = \bar{R},$$

where $\bar{R} \equiv \max\{\bar{R}^u, \bar{R}^b\}$ is given by **Lemma 2**.

Proof. From (11),

$$\beta E_t[\mathcal{M}_{t,t+1}^*(R_{f,t}^W - R_{p,t+1}^*)] = 0,$$

so

$$R_{f,t}^W = \frac{E_t[\mathcal{M}_{t,t+1}^* R_{p,t+1}^*]}{E_t[\mathcal{M}_{t,t+1}^*]}.$$

Since $\mathcal{M}_{t,t+1}^* > 0$ and, by **Lemma 2** we can let $\bar{R} \equiv \max\{\bar{R}^u, \bar{R}^b\}$,

$$R_{p,t+1}^* \leq \bar{R} \quad \text{a.s.,}$$

it follows that

$$R_{f,t}^W \leq \bar{R}.$$

Recall that $q_{US,t} > q_{W,t}$, hence

$$R_{f,t} = \frac{1}{q_{US,t}} < \frac{1}{q_{W,t}} = R_{f,t}^W \leq \bar{R}.$$

Therefore

$$R_{f,t} \leq \bar{R}_f, \quad \bar{R}_f \equiv \bar{R}.$$

□

As shown in (21), the convenience-yield term forces the US to be a net borrower:

$$\theta_t < 0, \quad 1 - \theta_t > 1.$$

Thus, for the US, only the left tail $\theta_t \downarrow -\infty$ is potentially problematic. The next lemma shows that the issuance cost prevents this from happening.

Lemma 4 (Endogenous lower bound on the US bond share). *Suppose the assumptions of Lemmas 2 and 3 hold, and suppose Assumption 1 holds. Then there exists a constant $\underline{\theta} > -\infty$ such that*

$$\theta_t \geq \underline{\theta} \quad \text{for all } t \leq \tau.$$

Proof. Fix t . After saving separation, the US chooses (ω_t, θ_t) to maximize

$$J_t(\omega, \theta) \equiv \frac{\beta}{1-\gamma} \log E_t [R_{A,t+1}(\omega, \theta)^{1-\gamma}] - \frac{\kappa}{2} (\omega - \bar{\omega})^2 - \eta \Upsilon(\theta),$$

where

$$R_{A,t+1}(\omega, \theta) = (1 - \theta)R_{p,t+1}(\omega) + \theta R_{f,t}, \quad R_{p,t+1}(\omega) = \omega R_{US,t+1} + (1 - \omega)R_{W,t+1}.$$

Consider the feasible reference choice $(\omega, \theta) = (\bar{\omega}, 0)$. By Lemma 2,

$$R_{p,t+1}(\bar{\omega}) \geq \underline{R} > 0,$$

so this choice yields a finite lower bound

$$J_t(\bar{\omega}, 0) \geq \underline{J} > -\infty,$$

where $\underline{J}$ is independent of t .

Now evaluate the objective at the equilibrium choice (ω_t, θ_t) . Since $\theta_t < 0$ by (21), and since Lemmas 2 and 3 give

$$R_{p,t+1}(\omega_t) \leq \bar{R}, \quad R_{f,t} \leq \bar{R}_f,$$

we have

$$R_{A,t+1}(\omega_t, \theta_t) = (1 - \theta_t)R_{p,t+1}(\omega_t) + \theta_t R_{f,t} \leq (1 - \theta_t)\bar{R} \leq \bar{R}(1 + |\theta_t|).$$

Hence

$$\frac{\beta}{1-\gamma} \log E_t[R_{A,t+1}(\omega_t, \theta_t)^{1-\gamma}] \leq \beta \log(\bar{R}(1 + |\theta_t|)),$$

and therefore

$$J_t(\omega_t, \theta_t) \leq \beta \log(\bar{R}(1 + |\theta_t|)) - \eta \Upsilon(\theta_t).$$

Since (ω_t, θ_t) is optimal,

$$J_t(\omega_t, \theta_t) \geq J_t(\bar{\omega}, 0) \geq \underline{J}.$$

Thus

$$\beta \log(\bar{R}(1 + |\theta_t|)) - \eta \Upsilon(\theta_t) \geq \underline{J}.$$

By **Assumption 1**,

$$\lim_{\theta \downarrow -\infty} \Upsilon'(\theta) = -\infty,$$

so, in particular,

$$\Upsilon(\theta) \rightarrow +\infty \quad \text{as } \theta \downarrow -\infty,$$

and $\Upsilon(\theta)$ dominates $\log(1 + |\theta|)$ in the left tail. Hence the last inequality cannot hold if $\theta_t \downarrow -\infty$. Therefore there exists $\underline{\theta} > -\infty$ such that

$$\theta_t \geq \underline{\theta} \quad \text{for all } t \leq \tau.$$

□

Assumption 10 (Positive lower bound on US total-saving returns).

$$(1 - \underline{\theta})\underline{R}_t + \underline{\theta} \bar{R}_f > 0,$$

where $\underline{\theta}$ is given by **Lemma 4**, and $\underline{R}_t \equiv \min\{\underline{R}^u, \underline{R}_t^b\}$, $\bar{R}_f$ are given by **Lemmas 2 and 3**.

Lemma 5 (Equilibrium bounds on total-saving returns and old-age consumption).

Suppose **Assumptions 1 and 6 to 10** hold. Then there exist constants $0 < \bar{R}_A^z < \infty$ and $0 < \underline{\lambda}_C^z \leq \bar{\lambda}_C^z < \infty$ such that, for each $z \in \{u, b\}$ and all $t \leq \tau$,

$$0 < R_{A,t}^z \leq \bar{R}_A^z < \infty,$$

and

$$\underline{\lambda}_C^z e_{US,t}^z \leq C_t^z \leq \bar{\lambda}_C^z e_{US,t}^z.$$

Proof. **Lemmas 2 to 4** imply that there exist constants $0 < \underline{R}^u \leq \bar{R}^u < \infty$, $\bar{R}^b < \infty$, $\bar{R}_f < \infty$, and $\underline{\theta} > -\infty$ together with $\underline{R}_t^b > 0$ such that

$$\underline{R}^u \leq R_{p,t}^u \leq \bar{R}^u, \quad \underline{R}_t^b \leq R_{p,t}^z \leq \bar{R}^b, \quad R_{f,t} \leq \bar{R}_f, \quad \theta_t \geq \underline{\theta}, \quad z \in \{u, b\}, \quad \forall t \leq \tau.$$

Also, by (21),

$$\theta_t < 0 \quad \text{for all } t \leq \tau.$$

For $z \in \{u, b\}$,

$$R_{A,t}^z = (1 - \theta_{t-1}^u) R_{p,t}^z + \theta_{t-1}^u R_{f,t-1}.$$

Using

$$R_{p,t}^u \geq \underline{R}^u, \quad R_{p,t}^b \geq \underline{R}_t^b, \quad R_{f,t-1} \leq \bar{R}_f, \quad \theta_{t-1}^u \in [\underline{\theta}, 0),$$

we obtain

$$\begin{aligned} R_{A,t}^u &\geq (1 - \theta_{t-1}^u) \underline{R}^u + \theta_{t-1}^u \bar{R}_f \geq \min\{\underline{R}^u, (1 - \underline{\theta}) \underline{R}^u + \underline{\theta} \bar{R}_f\} \equiv \underline{R}_A^u, \\ R_{A,t}^b &\geq (1 - \theta_{t-1}^u) \underline{R}_t^b + \theta_{t-1}^u \bar{R}_f \geq \min\{\underline{R}_t^b, (1 - \underline{\theta}) \underline{R}_t^b + \underline{\theta} \bar{R}_f\} \equiv \underline{R}_{A,t}^b. \end{aligned}$$

By **Assumption 10**, $\underline{R}_A^u, \underline{R}_{A,t}^b > 0$.

Similarly,

$$R_{A,t}^z \leq (1 - \theta_{t-1}^u) \bar{R}^z + \theta_{t-1}^u R_{f,t-1} \leq (1 - \theta_{t-1}^u) \bar{R}^z \leq (1 - \underline{\theta}) \bar{R}^z \equiv \bar{R}_A^z.$$

Therefore,

$$\begin{aligned} 0 &< \underline{R}_A^u \leq R_{A,t}^u \leq \bar{R}_A^u < \infty, \\ 0 &< \underline{R}_{A,t}^b \leq R_{A,t}^b \leq \bar{R}_A^b < \infty. \end{aligned}$$

Since

$$C_t^z = A_{t-1}^u R_{A,t}^z = \beta e_{US,t-1}^u R_{A,t}^z,$$

and by **Assumption 9** it follows that

$$\frac{\beta \underline{R}_A^u}{\underline{g}_u} e_{US,t}^u \leq \beta e_{US,t-1}^u \underline{R}_A^u \leq C_t^u \leq \beta e_{US,t-1}^u \bar{R}_A^u \leq \frac{\beta \bar{R}_A^u}{\underline{g}_u} e_{US,t}^u.$$

For C_t^b , we first show that its upper bound is proportional to e_t^b :

$$\begin{aligned} C_t^b &= \beta e_{US,t-1}^u R_{A,t}^b \\ &\leq \beta e_{US,t-1}^u (1 - \underline{\theta}) R_{p,t}^b \\ &\leq \beta e_{US,t-1}^u (1 - \underline{\theta}) \frac{(\bar{q} + \bar{d}) e_{US,t}^b}{\underline{q} e_{US,t-1}^u} \\ &= \beta (1 - \underline{\theta}) \frac{(\bar{q} + \bar{d})}{\underline{q}} e_{US,t}^b \end{aligned}$$

For the lower bound, we first consider the case with $\underline{R}_{A,t}^b = (1 - \underline{\theta})\underline{R}_t^b + \underline{\theta}\bar{R}_f$:

$$\begin{aligned}
C_t^b &= \beta e_{US,t-1}^u R_{A,t}^b \\
&\geq \beta e_{US,t-1}^u [(1 - \underline{\theta})\underline{R}_t^b + \underline{\theta}\bar{R}_f] \\
&= \beta e_{US,t-1}^u \left[(1 - \underline{\theta}) \frac{q e_{US,t}^b}{\bar{q} e_{US,t-1}^u} + \underline{\theta}\bar{R}_f \right] \\
&= \beta(1 - \underline{\theta}) \frac{q}{\bar{q}} e_{US,t}^b + \beta \frac{e_{US,t-1}^u}{e_{US,t}^u} \frac{e_{US,t}^u}{e_{US,t}^b} e_{US,t}^b \underline{\theta}\bar{R}_f \\
&\geq \left[\beta(1 - \underline{\theta}) \frac{q}{\bar{q}} + \beta \frac{\bar{\lambda}_e}{\underline{g}_u} \underline{\theta}\bar{R}_f \right] e_{US,t}^b;
\end{aligned}$$

if $\underline{R}_{A,t}^b = \underline{R}_t^b$, then

$$C_t^b \geq \beta e_{US,t-1}^u \underline{R}_t^b = \beta e_{US,t-1}^u \frac{q e_{US,t}^b}{\bar{q} e_{US,t-1}^u} = \beta \frac{q}{\bar{q}} e_{US,t}^b.$$

Hence one may take

$$\begin{aligned}
\underline{\lambda}_C^u &\equiv \frac{\beta \underline{R}_A^u}{\underline{g}_u} > 0, & \bar{\lambda}_C^u &\equiv \frac{\beta \bar{R}_A^u}{\underline{g}_u} < \infty, \\
\underline{\lambda}_C^b &\equiv \min\left\{ \beta \frac{q}{\bar{q}}, \beta(1 - \underline{\theta}) \frac{q}{\bar{q}} + \beta \frac{\bar{\lambda}_e}{\underline{g}_u} \underline{\theta}\bar{R}_f \right\} > 0, \\
\bar{\lambda}_C^b &\equiv \beta(1 - \underline{\theta}) \frac{(\bar{q} + \bar{d})}{\underline{q}} < \infty,
\end{aligned}$$

which proves the claim. $\square$

With these ingredients, we can state a single corollary that gives primitive conditions ensuring that the summability conditions in [Theorem 1](#) reduce to (26).

Corollary 1. Suppose [Assumptions 1](#) and [3](#) to [10](#) hold. For $z \in \{u, b\}$, let x_t^z be the value of x_t^z conditional on $z_0 = \dots = z_{t-1} = u$ and $z_t = z$. Then the two conditions (23) in [Theorem 1](#) are equivalent to the primitive conditions

$$\sum_{t=1}^{\infty} \frac{D_t^u}{e_t^u} < \infty, \tag{26a}$$

$$\sum_{t=1}^{\infty} \left(\frac{e_t^b}{e_t^u} \right)^{1-\gamma} < \infty. \tag{26b}$$

Proof. We show (a) (23a) $\iff$ (26a) and (b) (23b) $\iff$ (26b).

(a) Proof of (23a) $\iff$ (26a). By [Lemma 1](#), there exist constants $0 < \underline{q} \leq \bar{q} < \infty$ such

that

$$\underline{q} e_t^u \leq Q_t^u \leq \bar{q} e_t^u \quad \text{for all } t \leq \tau.$$

Therefore

$$\sum_{t=1}^{\infty} \frac{D_t^u}{Q_t^u} < \infty \iff \sum_{t=1}^{\infty} \frac{D_t^u}{e_t^u} < \infty.$$

(b) Proof of (23b) $\iff$ (26b).

By Lemma 1,

$$\underline{q} e_t^z \leq Q_t^z \leq \bar{q} e_t^z \quad \text{for all } t \leq \tau.$$

By Assumption 9, there exist constants $0 < \underline{q}_b \leq \bar{q}_b < \infty$ such that

$$\underline{q}_b e_t^b \leq Q_t^b + D_t^b \leq \bar{q}_b e_t^b \quad \text{for all } t \leq \tau,$$

where one may take

$$\underline{q}_b \equiv \underline{q}, \quad \bar{q}_b \equiv \bar{q} + \bar{\lambda}_{D_{US}}.$$

By Lemma 5, there exist constants $0 < \underline{\lambda}_C^z \leq \bar{\lambda}_C^z < \infty$ such that

$$\underline{\lambda}_C^z e_t^z \leq C_t^z \leq \bar{\lambda}_C^z e_t^z, \quad z \in \{u, b\}, \quad \forall t \leq \tau.$$

Hence

$$\frac{Q_t^b + D_t^b}{(C_t^b)^\gamma} \propto \frac{e_t^b}{(e_t^b)^\gamma} = (e_t^b)^{1-\gamma}, \quad \frac{Q_t^u}{(C_t^u)^\gamma} \propto \frac{e_t^u}{(e_t^u)^\gamma} = (e_t^u)^{1-\gamma},$$

This proves (23b) $\iff$ (26b). □

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